# A Statement With Some Shape

A synthetic example

**Theorem 1.** *Let $E$ be a nonempty type and $1 \le j \le k$. For every $x$ there is a constant $C$ such that, for every $n$, both $x \le C$ and $n + x \le n + C$.*

**Theorem 2.** *For every $j$ there is a function $x$ with $x(j) = j$ and $j \le x(j)$, and there is a boolean $x$ that is true.*

**Theorem 3.** *Let $E$ be nonempty. No natural number is negative, and either every $n$ satisfies $0 \le n$ or, if $1 = 1$, some $m$ equals $1$.*

**Definition 1.** *A number $n$ is a multiple of $m$ above $b$ if $n = mq$ for some $q \ge b$. The Fibonacci numbers are $F_0 = 0$, $F_1 = 1$ and $F_{n+2} = F_n + F_{n+1}$.*

**Theorem 4.** *Let $s$ be a solution for $d$ with bound at most $5$. Then $s$ starts at most at $5$.*

**Theorem 5.** *If every admissible datum has a good solution, then $0$ has a good solution.*

**Theorem 6.** *Every admissible datum $d$ has a good solution.*

**Theorem 7.** *Every positive number is at least $1$.*

**Theorem 8.** *A type with a single element has at most one element, and exactly one.*

**Theorem 9.** *On a probability space, the whole space has measure one.*

**Theorem 10.** *No solution for $0$ has a negative bound.*

**Theorem 11.** *Two solutions $s$ and $t$ for the same datum start at the same point.*

**Theorem 12.** *If $a \le b$ and $b \le a$, then $a = b$.*

**Definition 2.** *A profile is a function $f$ with $f(n) \le C$ for every $n$. A concrete profile is a profile that vanishes from $N$ on and has period $p$ below $N$.*

**Theorem 13.** *The zero profile below $N$ is a concrete profile, and $f(n + 1) = f(n)$ for it.*

**Theorem 14.** *A concrete profile vanishes from $N$ on.*

**Theorem 15.** *A scaled family for $d$ and $w$ that starts at $d$ starts at $d$.*

**Theorem 16.** *An object with both extents has a positive first coordinate.*

**Theorem 17.** *If the first $n + 5$ differences of $f$ are bounded by $C$ on $S$, so are the first $n + 4$.*

**Theorem 18.** *A vector bounded by $i$ is bounded by $i + 1$; vectors bounded by $7$ are bounded by $8$.*

**Theorem 19.** *The number of steps down from $k + 3$ is positive, and* $\operatorname{pick}(k + 1, 0) = 1$.

**Theorem 20.** *If every $x$ with $A(x)$ has some $y$ with $B(x, y)$, and $A(a)$ holds, then some $y$ has $B(a, y)$.*

**Theorem 21.** *For every $x$ there is $y$ such that $x + z \leq y + z$ for every $z$.*

**Theorem 22.** *If $P$ holds, some $x$ is $0$; $Q$ holds exactly when some $n$ equals itself; and either some $x$ is $1$ or $Q$ holds.*

**Theorem 23.** *Some map $\Phi \colon G \to H$ over $\mathbb{N}$ has both $A$ and $B$.*

**Theorem 24.** *There is a constant $C$ with $n \leq Cn^2 + n$ for every $n$; and for every $n$ there is such a constant.*

**Theorem 25.** *Let $\mu$ and $\nu$ be $\sigma$-finite measures and $\Phi \colon L^2(\mu \otimes \nu) \to \mathbb{R}^3$ a linear isometry. Every class $F$ has a representative, unique almost everywhere.*

**Theorem 26.** *If $m \leq b$ for every $m \leq n$, then some real $x$ exceeds $n$, or $n = 2x$ for some $x$.*

**Theorem 27.** *Let $m$ be a $\sigma$-algebra on $\Omega$ besides the given one. If $f$ is strongly measurable and $g$ is measurable for $m$, and $g$ is strongly measurable for the given one, then $g$ is strongly measurable for $m$.*

**Definition 3.** *The value of the digit $d_i$ is $i$, for $i = 0, \ldots, 69$.*

**Theorem 28.** *The value of the digit $d_{69}$ is $69$.*

**Theorem 29.** *A digit whose constructor index is $69$ has the value $69$.*

**Theorem 30.** *Half of $k + 2$ is less than $k + 2 + k$.*